

## Integration by Substitution

Some times we don't integrate straight forward. Then we use substitute to other variable, and we call sometimes u-substitution method. When we have a composite function and we need to integrate it then we use substitution method.

Suppose  $F(g(x))$  is a composite function.

$$\frac{d}{dx}[F(g(x))] = F'(g(x)) g'(x)$$

$$\int F'(g(x)) g'(x) dx = F(g(x)) + C$$

Since  $F$  is an antiderivative of  $f$  so we write

$$\int \underline{f(g(x))} \underline{g'(x) dx} = F(g(x)) + C$$

Here we say -  $u = g(x)$

$$\frac{du}{dx} = g'(x)$$

$$du = g'(x) dx$$

which implies

$$\int f(u) du = F(u) + C.$$

Problem: 1. Evaluate  $\int 3(8y-1)e^{4y^2-y} dy$

Say  $u = 4y^2 - y$   $\frac{du}{dy} = g'(y)$

$$\frac{du}{dy} = 8y - 1$$

$$du = (8y-1)dy$$

Now,  $\int 3e^{4y^2-y}(8y-1)dy$

$$= \int 3e^u du = 3 \int e^u du = 3e^u + C$$
$$= 3e^{4y^2-y} + C$$

2. Evaluate:  $\int x^2(3-10x^3)^4 dx$

Say,  $u = 3 - 10x^3$

$$\frac{du}{dx} = -30x^2$$

$$du = -30x^2 dx$$

$$-\frac{1}{30} du = x^2 dx$$

$$\int x^2(3-10x^3)^4 dx = \int u^4 \left(-\frac{1}{30}\right) du$$
$$= -\frac{1}{30} \int u^4 du$$
$$= -\frac{1}{30} \frac{u^5}{5} + C$$
$$= -\frac{1}{150} (3-10x^3)^5 + C$$

3. Evaluate:  $\int \cos 5x dx$

Why we substitute  $u = 5x$  (?) because ~~this integrand~~  
we integrate this integrand w.r.to  $x$  NOT  $5x$ .

so  $u = 5x$

$$\frac{du}{dx} = 5$$

$$du = 5 dx \Rightarrow \frac{1}{5} du = dx$$

$$\begin{aligned} \int \cos 5x dx &= \int \cos u \cdot \frac{1}{5} du = \frac{1}{5} \int \cos u du \\ &= \frac{1}{5} \sin u + C \\ &= \frac{1}{5} \sin 5x + C \end{aligned}$$

✗

4. Evaluate:  $\int \sin^3 x \cos x dx$

let  $u = \sin x$

$$\frac{du}{dx} = \cos x$$

$$du = \cos x dx$$

$$\int \sin^3 x \cos x dx = \int u^2 du = \frac{u^3}{3} + C = \frac{1}{3} \sin^3 x + C$$

✗

### Problems:

1. Evaluate:  $\int \frac{e^x}{\sqrt{1-e^{2x}}} dx$

Say,  $e^x = u$

$$e^x dx = du$$

$$\therefore \int \frac{e^x}{\sqrt{1-(e^x)^2}} dx = \int \frac{du}{\sqrt{1-u^2}} = \sin^{-1} u + C = \sin^{-1}(e^x) + C \quad \times$$

2.  $\int \cos^3 x dx = \int \cos x \cos^2 x dx = \int (1 - \sin^2 x) \cos x dx$

Say  $\sin x = u$

$$\cos x dx = du$$

$$\therefore \int \cos^3 x dx = \int (1 - u^2) du = u - \frac{u^3}{3} + C = \sin x - \frac{1}{3} \sin^3 x + C \quad \times$$

### Extra Problem:

Exercise: 5.3

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PROBLEMS: 15-56 [Odd numbers have answer in the book: see last page]

Exercise: Evaluate: the following integrals.

1.  $\int t^4 \sqrt{3-5t^5} dt$

2.  $\int \frac{e^x}{\sqrt{1-e^{2x}}} dx$

3.  $\int t \sqrt{7t^2+12} dt$

4.  $\int \frac{x^2+1}{\sqrt{x^2+3x}} dx$

5.  $\int e^{\sin x} \cos x dx$

6.  $\int \frac{e^x + e^{-x}}{e^x - e^{-x}} dx$

7.  $\int \frac{\sec^2 x}{\sqrt{1-\tan^2 x}} dx$

8.  $\int \frac{3^x}{(3^x+1)^2} dx$

— x —

For example, 1.  $\frac{d}{dx} \int_1^x \sin t \, dt = \sin(x)$  \*

2.  $\frac{d}{dx} \int_0^x \frac{dt}{1+\sqrt{t}} = \frac{1}{1+\sqrt{x}}$  \*

PROBLEMS: BOOK — H. Anton [Reference Book]

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PROBLEMS: 59-62.

Q1. Define  $F(x)$  by  $F(x) = \int_1^x (3t^2 - 3) \, dt$ . Find  $F'(x)$ .

Q2. Define  $F(t)$  by  $F(t) = \int_{\pi/4}^t \cos 2x \, dx$ . Find  $F'(t)$ .

## Integration by Parts

Our primary goal is how to integrate product of two functions.

For example,

$$\int f(x) g(x) dx$$

Here we use a formula to integrate product of two functions which we call it integration by parts.

Say,  $u = f(x) \quad du = f'(x) dx$

$$g(x) dx = dv$$

$$\boxed{\int u dv = uv - \int v du}$$

$$\int u dv + \int v du = uv$$

Example: Evaluate:  $\int x \cos x dx$

Say,  $u = x$  and  $dv = \cos x dx$

$$du = dx$$

$$v = \sin x$$

$$\int x \cos x dx = x \sin x - \int \sin x dx$$

$$= x \sin x - (-\cos x) + C$$

$$= x \sin x + \cos x + C$$

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Example: Evaluate  $\int e^x \cos x \, dx$

Sol.

$$\begin{array}{l} \text{Say, } u = \cos x \\ du = -\sin x \, dx \end{array} \quad \left| \begin{array}{l} dv = e^x \, dx \\ v = e^x \end{array} \right.$$

L — Logarithmic  
I — Inverse trigonometric  
A — Algebraic  
T — Trigonometric  
E — Exponential

Thus,

$$\int e^x \cos x \, dx = e^x \cos x - \int e^x (-\sin x) \, dx$$

$$\therefore I = e^x \cos x + \int e^x \sin x \, dx \quad \text{--- (1)}$$

∅ Say,  $I = \int e^x \cos x \, dx$

$$\begin{array}{l} \text{Now let } \sin x = u \\ \cos x \, dx = du \end{array} \quad \begin{array}{l} dv = e^x \, dx \\ dv = e^x \end{array}$$

$$\int e^x \sin x \, dx = e^x \sin x - \int e^x \cos x \, dx = e^x \sin x - I$$

From (1)

$$I = e^x \cos x + e^x \sin x - I$$

$$2I = e^x \cos x + e^x \sin x$$

$$I = \frac{e^x}{2} (\cos x + \sin x)$$

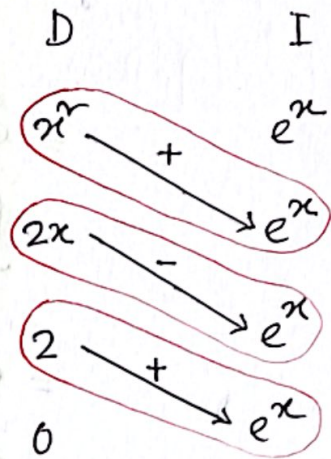
$$\text{Therefore } \int e^x \cos x \, dx = \frac{1}{2} e^x (\sin x + \cos x) + C$$

✱

We can also use shortcut techniques for ~~pro~~ integration by parts.

Example: Evaluate:  $\int x^3 \cdot e^x dx$

Sol<sup>n</sup>:  $\int x^3 e^x dx = x^3 e^x - 2x^2 e^x + 2x e^x + e^x + C$

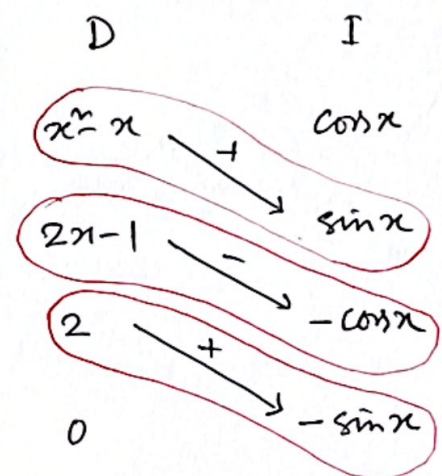


Example: Evaluate:  $\int (x^2 - x) \cos x dx$

Sol<sup>n</sup>:  $\int (x^2 - x) \cos x dx$

$$= (x^2 - x) \sin x + (2x - 1) (-) (-) \cos x + 2 (+) \cdot (- \sin x)$$

$$= (x^2 - x) \sin x + (2x - 1) \cos x - 2 \sin x + C$$



Multiply under inside red color's function with sign and use '+' inbetween two red bounded red color's mark.

Example: Evaluate  $\int e^x \sin x dx$

Sol:

Say,

$$I = \int e^x \sin x dx$$

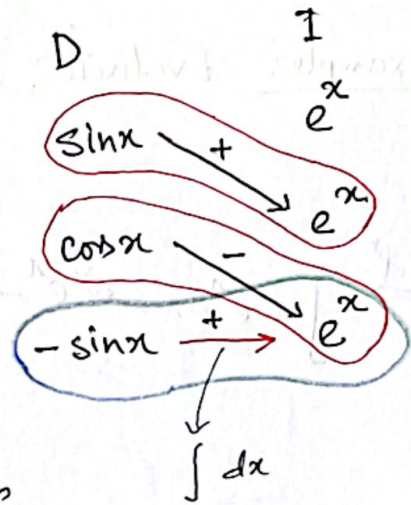
$$= e^x \sin x - e^x \cos x$$

$$- \int e^x \sin x dx$$

$$= e^x \sin x - e^x \cos x - I$$

$$\Rightarrow 2I = e^x \sin x - e^x \cos x + C$$

$$I = \frac{1}{2} e^x [\sin x - \cos x] + C$$



TRY YOURSELF ⑦ Evaluate:

1.  $\int x \ln x dx$

2.  $\int e^{3x} \cos 2x dx$

3.  $\int x \tan^{-1} x dx$

4.  $\int (x + x \cos x) dx$

5.  $\int x^3 e^{x^2} dx$

EXTRA PROBLEMS:

Exercise : 7.2

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PROBLEMS: 1-38 (TRY ODD NUMBERS)